Radio Map-Based Beamforming Assisted With Reduced Pilots

Bin Yang, *Graduate Student Member, IEEE*, Wei Wang, *Senior Member, IEEE*, and Wei Zhang, *Fellow, IEEE*

***Abstract*—Radio map is a promising technology that connects the user equipment (UE) location and its channel state information (CSI). By applying a radio map, the beamforming vector can be generated based solely on the location of the UE, thereby significantly saving pilot effort. However, the effectiveness of radio map-based beamforming is influenced by several factors, such as positioning errors and channel dynamics. To improve the adaptability and performance of radio map-based beamforming, in this paper we consider integrating location information with reduced pilots and examine the trade-off between pilot overhead and beamforming performance. In particular, an end-to-end method for joint extrapolation, denoising and performing beamforming with reduced pilots is proposed. Subsequently, the beamforming vector obtained from reduced pilots is integrated with that generated by the radio map. Considering the overhead caused by pilots, the support vector machine (SVM) is applied to determine whether it is worth introducing pilots for integration. According to the numerical simulations, the proposed end-to-end method with reduced pilots is superior to that with full pilots in terms of spectral efficiency (SE) and can be improved by integrating with the radio map. In addition, the application of SVM can effectively identify the integration needs, further reducing unnecessary pilot effort and thereby increasing SE.**

***Index Terms*—Beamforming, deep learning, radio map, reduced pilots, SVM.**

I. Introduction

MULTIPLE-INPUT multiple-output (MIMO) [1], [2] brings much improvement to the communication performance of user equipment (UE) by equipping massive antennas at the base station (BS) side [3]. To fully leverage its benefits, MIMO relies on accurate instantaneous channel state information (CSI) of the downlink channel for beamforming [1]. Pilot-based methods [4] are widely used to acquire CSI in current wireless communications. For example, in the 3rd Generation Partnership Project (3GPP) standards for 5G [5], [6], CSI-reference signal (CSI-RS) estimates CSI according to the pilots inserted into the downlink channel to facilitate beamforming for MIMO transmission. However, pilot-based methods incur significant overhead with the increase in the number of antennas. To be specific, in frequency-division duplex (FDD) systems, the BS needs to transmit high-density pilots in the downlink for CSI estimation at the UE side, followed by a feedback mechanism from the UE [7]. This results in a substantial portion of time-frequency resources being dedicated to the pilots, thereby reducing the spectral efficiency (SE). In addition, pilots are also vulnerable to noise and interference [8], resulting in poor performance especially at low signal-to-noise ratios (SNRs).

In the recommendation for International Mobile Telecommunications (IMT)-2030 [9], integrated sensing and communication (ISAC), and integrated AI and communication are added into the usage scenarios of 6G. This suggests that 6G will possess sensing capabilities and is expected to leverage AI compared to 5G. In this context, the location information of UE can be obtained through the sensing capabilities of the network. Previously, after getting the CSI feedback from the UE, the BS would discard the obtained CSI once used. However, if the CSI is labeled with location information, it can be retained for further analysis. By utilizing the powerful tools of AI, the relationship between locations and CSI can be learned, which advances the research into radio map [10], [11], [12], [13]. Initially, radio map is defined as the geographical signal power spectrum density as a function of location, frequency, and time [10]. Then, it has been expanded to more channel-related information like pathloss, time of arrival (ToA), phase, azimuth, and zenith angle of departure (AoD), and angle of arrival (AoA) [11], [12], [13]. In [14], [15], and [16], radio map has been further customized to be task-oriented, capable of directly generating beamforming vector based on UE's location information. Owing to this end-to-end design and the advanced capabilities of deep neural network (DNN), the trained radio map can achieve satisfactory beamforming performance solely with UE's location. However, the effectiveness of radio map-based beamforming is influenced by multiple factors, such as positioning errors and channel dynamics, which significantly constrains its maximum potential.

As two distinct technologies, pilot-based beamforming and radio map-based beamforming have their unique advantages and shortcomings. Considering the characteristics of radio map and pilot-based methods, these two approaches are not mutually exclusive but can be integrated to complement each other, raising both the upper and lower limits of the performance simultaneously. Specifically, radio map can pro-

Received 5 August 2024; revised 11 January 2025; accepted 28 April 2025. Date of publication 21 May 2025; date of current version 14 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201309 and in part by China Scholarship Council under Grant 202106960008. The associate editor coordinating the review of this article and approving it for publication was N. Shlezinger. *(Corresponding author: Wei Wang.)*

Bin Yang and Wei Zhang are with the School of Electrical Engineering and Telecommunications, The University of New South Wales, Sydney, NSW 2052, Australia (e-mail: bin.yang@unsw.edu.au; w.zhang@unsw.edu.au).

Wei Wang is with the Peng Cheng Laboratory, Shenzhen 518055, China (e-mail: wangw01@pcl.ac.cn).

Data is available on-line at https://github.com/yangbin-xd/reduced-pilots.git.

Digital Object Identifier 10.1109/TWC.2025.3569567

1536-1276 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

vide prior information on location, thereby significantly reducing pilot overhead. The location information can also help to counteract the loss caused by pilot contamination, offering a robust means to enhance the stability of beamforming performance. Conversely, as real-time measurements of the channel, pilots can assist radio map in enhancing adaptability to positioning errors and channel dynamics. Besides, pilots can offer accurate channel measurements at high SNRs, improving the maximum performance of radio map.

A. Related Works and Motivations

Related works in this field adopt two principal strategies. One strategy focuses on reduced pilots [17], [18] such as compressive sensing (CS) methods [19], channel extrapolation [20] and some robust deep learning approaches [21], [22], which enhance SE through the reduction of pilot overhead. The other strategy incorporates location information to assist beamforming like geometry-based method (GBM) [23], [24], [25] and channel knowledge map (CKM) [11], [12], [13].

For the first strategy, reduced pilots will inevitably result in the challenge of solving an under-determined problem. Generally speaking, the number of pilot measurements should be no smaller than the number of transmit antennas to estimate all the channel coefficients [18]. In this case, CS methods [19] are introduced to transform the under-determined problem into an optimization problem for solution by utilizing the sparsity of channels. Classical CS-based channel estimation algorithms include the basis pursuit (BP) [26], subspace pursuit (SP) [27], and orthogonal matching pursuit (OMP) [28]. These CS-based algorithms are of prohibitively high complexity, and the underlying theory is established under some ideal assumptions, e.g., sparsity known a priori and discrete support [19]. The aforementioned challenges seriously degrade the applicability of CS-based methods. Channel extrapolation [20] is another approach for channel estimation with reduced pilots and can be performed over antenna dimension. To be specific, the BS sends pilots to the UE from just a subset of its antennas. After obtaining the channel gains of these antennas, the channel gains of the remaining antennas can be inferred by exploiting the spatial correlation. In [29], the channel gains of the antennas that do not transmit pilots are obtained directly by averaging the channel gains of their adjacent antennas. In [30], a parametrized model is employed to estimate the AoA, which is applied to compute the complete CSI. In [17], the minimum mean square error (MMSE) estimator is used to compute the channel coefficients at the antennas that do not transmit pilots. In addition, robust deep learning approaches are also applied to address the challenge of reduced pilots. For instance, pre-trained models for channel estimation are utilized in [31] and [32], and satisfactory channel estimation performance can be achieved with few pilots through transfer learning. In [33], pilots from previous transmissions are utilized to enable the model to learn demodulation with few pilots by meta-learning [21]. In [34], data augmentation [22] is also used to improve the channel estimation performance with few pilots.

For the second strategy, location information is introduced in GBM [23], [24], [25] and CKM [11], [12], [13] to mitigate the reliance on pilots. The main idea of GBM is to calculate the AoD from the BS to the UE based on their locations. Then, the beamforming vector can be designed with the obtained AoD to align the beam with the UE. The GBM works well under line-of-sight (LoS) conditions but deteriorates visibly in non-LoS (NLoS) situations. To address the issue of NLoS, CKM is built to estimate the AoA of BS. The CKM can be seen as a large database that stores the potential positions of UE and the corresponding AoAs. After getting the UE's location, k nearest neighbors (kNN) [35] and inverse distance weighting (IDW) [36] methods are applied to do interpolation for AoA. However, kNN algorithm requires distance calculation and sorting, which brings a substantial computational burden when the size of the measurement data is large. Moreover, the interpolation by IDW struggles to achieve satisfactory results in complex wireless environments.

B. Contributions and Paper Organization

Motivated by the above discussions, this paper utilizes reduced pilots to assist radio map-based beamforming. The integration strategy of radio map and reduced pilots goes beyond a simple combination, serving as a comprehensive solution that adapts to varying environments. The balance between pilot overhead and beamforming performance is examined in this integration. To be specific, this paper makes the following contributions.

- We propose a task-oriented design for beamforming with reduced pilots based on deep learning. The designed neural network can perform joint extrapolation, denoising, and beamforming with the reduced pilots. The trade-off between pilot overhead and beamforming performance is investigated.
- We study the performance characteristics of the beamforming schemes based on reduced pilots and radio map. To leverage the strengths and compensate for the weaknesses of these two approaches, we integrate the beamforming vectors obtained from the reduced pilots with those generated by radio map.
- Considering the pilot overhead, we apply support vector machine (SVM) to decide whether it is worthwhile to allocate and integrate pilots based on UE's location. This step aims to remove unnecessary pilot overhead and approach the optimal trade-off between pilot overhead and beamforming performance.

The numerical simulations verify the proposed schemes in this paper. Specifically, we validate the superiority of the reduced pilots-based beamforming empowered by deep learning compared to the conventional full pilot scheme. The integration of radio map and reduced pilots can effectively elevate both the performance floor and ceiling. Furthermore, the application of SVM can further reduce unnecessary pilot overhead, improving the downlink SE for data transmission. The proposed approach is also applicable in multi-UE scenarios by applying the zero-forcing (ZF) criterion.

The remainder of the paper is organized as follows. In Section II, the system model of radio map-based beamforming assisted with reduced pilots is presented. In Section III, the deep learning-based reduced pilots method for beamforming is proposed. In Section IV, an integration strategy of radio map

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

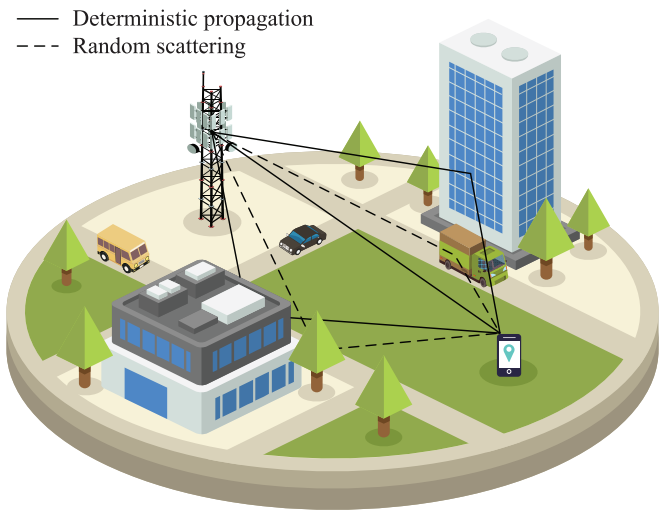


Fig. 1. The channel model of wireless communications with deterministic propagation and random scattering.

and reduced pilots has been proposed, with the decision to allocate and integrate pilots based on SVM. In Section V, the aforementioned proposal is extended to multi-UE scenarios. In Section VI, numerical simulations and results are given, followed by conclusions in Section VII.

Notations: a, $\mathbf{a}$ and $\mathbf{A}$ represent a scalar, a vector, and a matrix, respectively. $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$ and $(\cdot)^\dagger$ represent conjugate, transpose, conjugate transpose and Moore-Penrose inverse operations, respectively. $||\cdot||$ denotes the l_2 (Euclidean) norm. $\mathcal{CN}(\mu, \sigma^2)$ represents complex Gaussian distribution with the mean μ and the standard deviation σ. $\mathbb{R}$ and $\mathbb{C}$ stand for the set of real and complex numbers, respectively. $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of the complex number, respectively. $\odot$ and $\oslash$ denote the element-wise (Hadamard) product and division, respectively.

II. System Model

We consider a single-cell multi-input single-output (MISO)-orthogonal frequency division multiplexing (OFDM) downlink transmission scenario where a BS equipped with M transmit antennas serves a single-antenna UE over K subcarriers. This scenario for single UE is extended to multiple UEs in Section V. The number of samples for the UE at different locations is L, and the location of UE is available.

A. Channel Model

In wireless communications, the channel is determined by the propagation paths of electromagnetic waves. Typically, the wireless channel can be decomposed into two components. One is the deterministic propagation caused by buildings and terrain, which is primarily determined by the UE's location and surrounding environment. The other is the random scattering induced by dynamic factors, such as the movement of objects within the environment, including vehicles, pedestrians, and even wind-driven vegetation. The channel model of wireless communications with deterministic propagation and random scattering is illustrated in Fig. 1. The downlink channel matrix $\mathbf{H} \in \mathbb{C}^{K\times M}$ can be decomposed by

$$\mathbf{H} = \mathbf{H}^d + \mathbf{H}^r, \tag{1}$$

where $\mathbf{H}^d \in \mathbb{C}^{K\times M}$ represents the deterministic propagation channel, and $\mathbf{H}^r \in \mathbb{C}^{K\times M}$ models the random scattering in the environment. The ratio of the power of $\mathbf{H}^r$ to that of $\mathbf{H}^d$ is denoted by κ. We consider scenarios where deterministic propagation dominates, i.e. $\kappa < 1$.

B. Framework

According to the channel model, the dominant part $\mathbf{H}^d$ is related to UE's location and can thus be obtained from radio map. However, radio map-based beamforming cannot capture $\mathbf{H}^r$. Therefore, reduced pilots are incorporated to account for the effects of random scattering. The framework of radio map-based beamforming assisted with reduced pilots is shown in Fig. 2, which consists of four main parts.

- The red module is reduced pilots-based beamforming. In this module, the received reduced pilots $\boldsymbol{p}^\eta$ of the UE are given to the convolutional neural network (CNN), where the superscript η is the proportion of antennas transmitting pilots to the total number of antennas at the BS. Then the beamforming vector $\mathbf{V}^{\mathrm{P}}$ can be obtained by reduced pilots-based beamforming.
- The yellow module represents radio map-based beamforming, where the location $\boldsymbol{g}$ of the UE is input to the DNN to generate the beamforming vector $\mathbf{V}^{\mathrm{M}}$.
- The green module is the integration of reduced pilots and radio map, which combines the output of the aforementioned modules and inputs them into the CNN to get the integrated beamforming vector $\mathbf{V}^{\mathrm{I}}$.
- The blue module is SVM-based discriminator, which also takes $\boldsymbol{g}$ as input to the SVM to help the discriminator decide whether to adopt $\mathbf{V}^{\mathrm{M}}$ or $\mathbf{V}^{\mathrm{I}}$ as the final beamforming vector $\mathbf{V}^{\mathrm{SU}}$ for single UE.

In the practical implementation of this scheme, the modules for radio map-based beamforming and the SVM-based discriminator are initially executed once the BS has acquired the UE's location $\boldsymbol{g}$. If the discriminator decides to adopt $\mathbf{V}^{\mathrm{M}}$, there is no need for the BS to configure pilots in the downlink, thus reducing pilot overhead and saving operation time. However, if the discriminator opts for $\mathbf{V}^{\mathrm{I}}$, the BS will transmit reduced pilots to the UE. Next, reduced pilots-based beamforming is conducted and then fused with $\mathbf{V}^{\mathrm{M}}$ to obtain the integrated beamforming vector $\mathbf{V}^{\mathrm{I}}$.

C. Metric

The downlink SE for the data transmission is selected as the key metric for evaluating the effectiveness of the beamforming strategy. In our considered scenario, the received signal of the UE over the kth subcarrier can be written as

$$y_k = \mathbf{h}_k^H \mathbf{x}_k + w_k = \mathbf{h}_k^H \mathbf{v}_k s_k + w_k, \tag{2}$$

where $\mathbf{h}_k \in \mathbb{C}^{M\times 1}$ is the downlink CSI, $\mathbf{x}_k = \mathbf{v}_k s_k \in \mathbb{C}^{M\times 1}$ is the transmitted signal where $\mathbf{v}_k \in \mathbb{C}^{M\times 1}$ is the beamforming vector and s_k is a complex number representing the transmitted symbol, $w_k \sim \mathcal{CN}(0, \sigma^2)$ is the complex additive Gaussian noise. The received SNR has the form of

$$r_k = \frac{\left|\mathbf{h}_k^H \mathbf{v}_k\right|^2}{\sigma^2}. \tag{3}$$

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

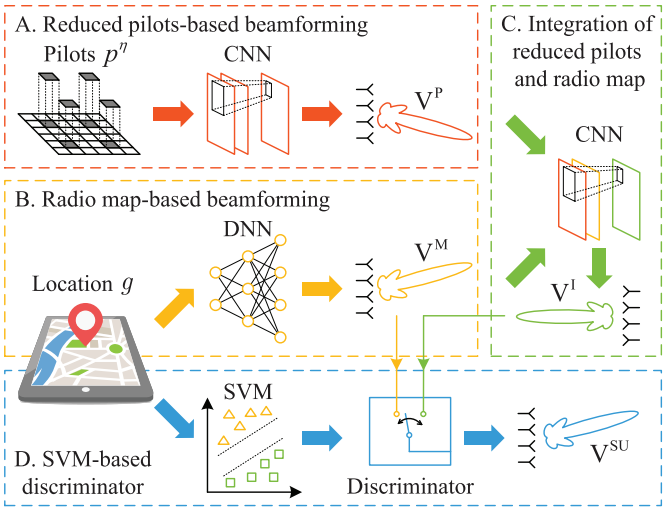


Fig. 2. The framework of radio map-based beamforming assisted with reduced pilots.

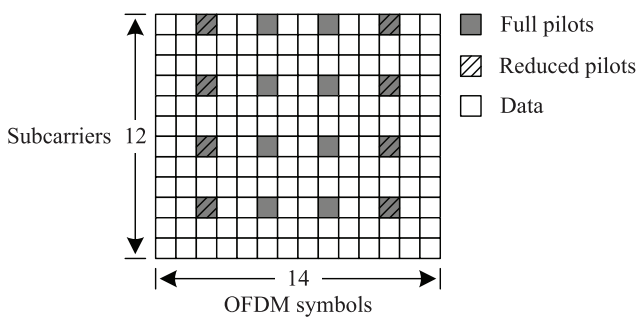


Fig. 3. An example of the pilot pattern in a resource block.

The SE for the data transmission of the UE over all subcarriers can be averaged to

$$R = \frac{1}{K}\sum_{k=1}^{K}\log\left(1+r_k\right)\left(1-\rho\right), \tag{4}$$

where ρ represents the proportion of the pilot overhead for the UE and will be further illustrated in Section III. A.

III. Reduced Pilots-Based Beamforming

In this section, we first introduce the concept of reduced pilots by illustrating the pilot pattern. Then, the reduced pilots-based beamforming approach is proposed.

A. Pilot Pattern

For the sake of discussion, we illustrate the pilot pattern within a resource block for a given slot in Fig. 3. In 3GPP technical specification (TS) 38.211 for release 18 [5], a resource block (RB) is defined as 12 consecutive subcarriers in the frequency domain, and a slot usually contains 14 OFDM symbols in the time domain. In this structure, a resource element (RE) is the smallest unit in the time-frequency domain, consisting of a single subcarrier and an OFDM symbol. During the downlink pilot transmission, all antennas need to send pilot symbols to estimate the channel gain of each antenna from the BS to the UE. To avoid interference among them, a combination of time division, frequency division, and code division multiplexing techniques is employed. Specifically, transmit antennas can occupy different time-frequency domains, i.e., REs for pilot transmission. Furthermore, to enhance the utilization of the time-frequency resources, multiple antennas can share the same RE by using orthogonal codes.

Taking $M = 32$ as an example for further illustration. We assume that every two antennas share the same RE for pilot transmission. As defined in Section II-A, η is the ratio of the number of antennas transmitting pilots to the total number of antennas at the BS. The setup where $\eta = 100\%$ is referred to as full pilots, meaning that every antenna at the BS transmits pilot symbols. In contrast, if $\eta < 100\%$, this configuration is referred to as reduced pilots, indicating that only a subset of the total antennas are used for pilot transmission. As shown in Fig. 3, 16 REs need to be allocated in full pilots setup, and only 8 REs are occupied for $\eta = 50\%$. The number of REs allocated for pilots to the total number of REs is denoted as ρ, which is mentioned in (4). In full pilots setup, $\rho = 16/(12\times 14) = 9.52\%$ while $\rho = 8/(12\times 14) = 4.76\%$ for $\eta = 50\%$ in reduced pilots setup.

B. Task-Oriented Design

The designed reduced pilots-based beamforming generates the beamforming vector $\mathbf{V}_l^{\mathrm{P}}$ of the UE at the lth location by taking its received pilots $\boldsymbol{p}_l^{\eta}$ as input, i.e.,

$$\mathbf{V}_l^{\mathrm{P}} = \mathcal{F}\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right), \tag{5}$$

where $\boldsymbol{p}_l^{\eta} = [Y_{l,1};Y_{l,2};\cdots;Y_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ and $Y_{l,i}$ is a complex number represents the received pilot symbol of the UE at the lth location, and θ_{P} is the set of the parameters of the neural network for reduced pilots-based beamforming.

Algorithm 1 Data Preprocessing

Input: $\boldsymbol{p}_l^{\eta} \in \mathbb{C}^{\eta M\times 1}$
Output: $\left[\Re\left(\mathbf{H}_l'\right),\Im\left(\mathbf{H}_l'\right)\right] \in \mathbb{C}^{K\times M\times 2}$
1: $\mathbf{h}_l^{\eta} \leftarrow \mathbf{p}_l^{\eta} \oslash \mathbf{x}_l^{\eta}$
2: $\mathbf{h}_l^{\eta} \leftarrow \mathbf{h}_l^{\eta}/||\mathbf{h}_l^{\eta}||$
3: $\mathbf{h}_l' \leftarrow$ zero-padding of $\mathbf{h}_l^{\eta}$
4: $\mathbf{H}_l' \leftarrow \left[\mathbf{h}_l',\mathbf{h}_l',\cdots,\mathbf{h}_l'\right]^H$
Return: $\left[\Re\left(\mathbf{H}_l'\right),\Im\left(\mathbf{H}_l'\right)\right]$

The architecture of the proposed reduced pilots-based beamforming is shown in Fig. 4. As shown in the figure, the received pilots need to be preprocessed to adapt to the input dimensions of the neural network. The first step involves channel estimation based on the received pilots. The channel gains of the antennas transmitting pilots to the UE can be obtained by least squares (LS) estimation [37] as

$$\mathbf{h}_l^{\eta} = \mathbf{p}_l^{\eta} \oslash \mathbf{x}_l^{\eta}, \tag{6}$$

where $\mathbf{h}_l^{\eta} = [h_{l,1};h_{l,2};\cdots;h_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ is the estimated channel gains of the antennas that transmitting pilots, $\mathbf{x}_l^{\eta} = [X_{l,1};X_{l,2};\cdots;X_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ represents the transmitted pilot symbols. Then, $\mathbf{h}_l^{\eta}$ is normalized to meet the constraint of $||\mathbf{h}_l^{\eta}|| = 1$. After that, zero-padding is performed on the gains of the remaining antennas to increase the dimension from ηM to M. The channel gains after padding with zeros

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

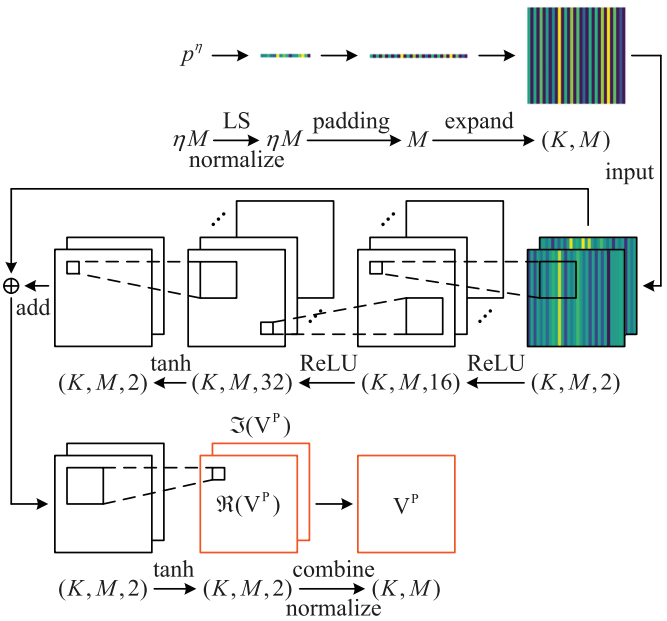


Fig. 4. The architecture of reduced pilots-based beamforming.

is denoted as $\mathbf{h}_l' \in \mathbb{C}^{M\times 1}$, which can be further extended to $\mathbf{H}_l' = \left[\mathbf{h}_l', \mathbf{h}_l', \cdots, \mathbf{h}_l'\right]^H \in \mathbb{C}^{K\times M}$ by repeating K times of $\mathbf{h}_l'$. The matrix $\mathbf{H}_l'$ can be seen as the partial CSI, and joint channel estimation and beamforming need to be performed for the beamforming vector. Finally, $\mathbf{H}_l'$ is decomposed into real and imaginary parts before being input into the neural network. The procedure of the data preprocessing is summarized in Algorithm 1.

Then, the task of the neural network is to generate the beamforming vector by jointly performing extrapolation, denoising, and beamforming with the preprocessed data. CNN is selected due to its success in image super-resolution [38]. In Fig. 4, the preprocessed data with dimensions $(K, M, 2)$ is fed into the neural network. The convolutional kernel size of the CNN is 3×3, which is a standard size in the field of image processing. The stride and zero-padding are both set to 1 to ensure that the output size remains unchanged. Two convolutional layers with rectified linear unit (ReLU) [39] activation function are employed to gradually increase the depth from 2 to 32 to extract more features of the input data. After that, these high-level features are aggregated to match the dimensions expected by the final output data, and hyperbolic tangent (tanh) is applied as the activation function. To enhance the performance of the neural network, additional layers can be added on this basis, and residual network (ResNet) [40] can be applied to prevent the degradation problem. As shown in Fig. 4, a residual connection is introduced to add the input to the output. This connection allows gradients to flow directly through several layers, ensuring that sufficient gradient reaches every layer of the network, which makes the training process more feasible and stable. After the addition operation, the last convolutional layer serves to map the output range to $(-1, 1)$ by utilizing the tanh activation function. Then, two channels of the tensor are combined into complex data with the dimension of (K, M), followed by normalization to obtain the generated beamforming vector $\mathbf{V}^{\mathrm{P}}$.

To train the built neural network, the loss function is customized to be the negative of the downlink SE in the scenario of single UE without taking interference into account as in [15].

$$\begin{aligned}
&\mathcal{L}\Big(\mathcal{F}\left(\boldsymbol{p}_l^\eta; \theta_{\mathrm{P}}\right), \mathbf{H}_l\Big) \\
&= -\mathcal{R}\Big(\mathcal{F}\left(\boldsymbol{p}_l^\eta; \theta_{\mathrm{P}}\right), \mathbf{H}_l\Big) \\
&= -\frac{1}{K}\sum_{k=1}^{K}\log\left(1 + \frac{\left|\mathbf{h}_{l,k}^H \mathcal{F}_k\left(\boldsymbol{p}_l^\eta; \theta_{\mathrm{P}}\right)\right|^2}{\sigma^2}\right),
\end{aligned} \tag{7}$$

where $\mathbf{H}_l = \left[\mathbf{h}_{l,1}, \mathbf{h}_{l,2}, \cdots, \mathbf{h}_{l,K}\right]^H \in \mathbb{C}^{K\times M}$ is the downlink CSI for the UE at the lth location and $\mathbf{h}_{l,k} \in \mathbb{C}^{M\times 1}$ denotes the channel gain of the kth subcarrier, $\mathcal{F}_k\left(\boldsymbol{p}_l^\eta; \theta_{\mathrm{P}}\right) \in \mathbb{C}^{M\times 1}$ is the generated beamforming vector over the kth subcarrier. The training objective is to minimize the expected loss function of the UE at all L locations, i.e.,

$$\theta_{\mathrm{P}}^* = \arg\min_{\theta_{\mathrm{P}}} \frac{1}{L}\sum_{l=1}^{L}\mathcal{L}\Big(\mathcal{F}\left(\boldsymbol{p}_l^\eta; \theta_{\mathrm{P}}\right), \mathbf{H}_l\Big). \tag{8}$$

It has been proved in [15] that the customized loss function in (7) is differentiable with respect to θ_{P}, which provides feasibility for gradient calculation and weight update.

IV. The Integration of Radio Map and Reduced Pilots

Pilot and location are two different data sources, leading to different performances of the beamforming schemes based on them. To be specific, pilot information can be obtained via CSI feedback by the UE, which is larger in size but more susceptible to noise and interference, while location information comes from global positioning system (GPS) or differential GPS (DGPS) [41], which is precise but has lower dimensions. Therefore, pilot-based beamforming approach achieves nearly optimal performance at high SNR but struggles at low SNR. In contrast, radio map-based beamforming scheme offers robust performance with a lower maximum potential. To leverage the strengths and compensate for the weaknesses of these two methods, we integrate location with reduced pilots for better performance in varying environments. Considering the effect of pilot overhead on SE, we further designed an SVM-based discriminator, which determines whether it is worth sparing some time-frequency resources on pilots to achieve a final performance improvement according to the UE's location.

A. Radio Map-Based Beamforming

Radio map establishes a connection between UE's location and its CSI, and radio map-based beamforming generates the beamforming vector $\mathbf{V}_l^{\mathrm{M}}$ for the UE by taking its location $\boldsymbol{g}_l$ as input, i.e.,

$$\mathbf{V}_l^{\mathrm{M}} = \mathcal{G}\left(\boldsymbol{g}_l; \theta_{\mathrm{M}}\right), \tag{9}$$

where $\boldsymbol{g}_l = [x_l, y_l]^T$ represents the two-dimensional coordinates of the UE at the lth location, and θ_{M} is the set of the parameters of the radio map.

The architecture of radio map-based beamforming is shown in Fig. 5. DNN is adopted which is proved to be able

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

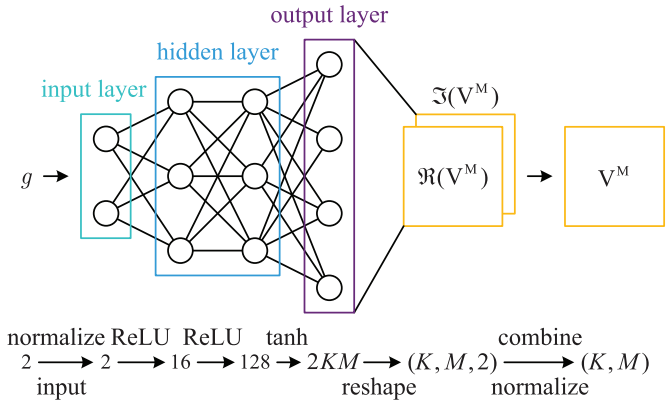


Fig. 5. The architecture of radio map-based beamforming.

to approximate any function that has continuous values by universal approximation theory [42]. As shown in the figure, the normalized location of UE with the range of $(0,1)$ is fed to the input layer, and ReLU is applied here to activate neurons, introducing non-linearity to the model without significantly affecting the computational complexity. Then, multiple hidden layers are utilized to expand the dimension of the input data and learn the relationship between the position and the beamforming vector. Next, the output layer applies tanh activation function to scale the output to $(-1,1)$ range. After that, the result of the output layer can be reshaped to obtain the real and imaginary parts of the beamforming vector, which can be combined to form the complex beamforming vector. Finally, normalization is performed to meet the power constraint.

During the training process, the same loss function for the reduced pilots is used, and the parameters can be obtained by minimizing the loss function defined in (7), i.e.,

$$\theta_{\mathrm{M}}^{*} = \arg\min_{\theta_{\mathrm{M}}} \frac{1}{L}\sum_{l=1}^{L} \mathcal{L}\Big(\mathcal{G}\left(\boldsymbol{g}_l;\theta_{\mathrm{M}}\right), \mathbf{H}_l^{d}\Big). \tag{10}$$

B. Integration Scheme

The integrated beamforming vector $\mathbf{V}_l^{\mathrm{I}}$ of the UE at the lth location is acquired by fusing the beamforming vectors $\mathbf{V}_l^{\mathrm{M}}$ and $\mathbf{V}_l^{\mathrm{P}}$ generated by radio map and reduced pilots, respectively, i.e.,

$$\mathbf{V}_l^{\mathrm{I}} = \mathcal{H}\left(\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{V}_l^{\mathrm{P}}\right);\theta_{\mathrm{I}}\right), \tag{11}$$

where θ_{I} is the set of the parameters of the neural network for integration.

The architecture of the integration scheme is shown in Fig. 6. In the figure, the beamforming vector V^{M} generated by the radio map and V^{P} obtained from reduced pilots are decomposed into real and imaginary parts. These decomposed parts are then concatenated to form a tensor of shape $(K,M,4)$, which contains high-dimensional representations of location and pilot information. CNN is employed with a convolutional kernel size of 3×3, and both stride and zero-padding are set to 1 to ensure the size remains unchanged. Similar to the structure in the reduced pilots-based beamforming as shown in Fig. 4, two convolutional layers are used to extract more features by increasing the volume from $(K,M,4)$ to $(K,M,32)$. ReLU is adopted as the activation function here. Subsequently, these features are aggregated to a $(K,M,2)$ tensor, and tanh is used to make the output range within $(-1,1)$. After that, two channels of the tensor are combined for the complex value, and normalization is performed to obtain the integrated beamforming vector V^{I}.

In the training part, the parameters for the integration scheme can be yielded by minimizing the loss function as

$$\theta_{\mathrm{I}}^{*} = \arg\min_{\theta_{\mathrm{I}}} \frac{1}{L}\sum_{l=1}^{L} \mathcal{L}\Big(\mathcal{H}\left(\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{V}_l^{\mathrm{P}}\right);\theta_{\mathrm{I}}\right), \mathbf{H}_l\Big), \tag{12}$$

C. SVM-Based Discriminator

When reduced pilots are introduced to assist the radio map, the integration scheme can improve the beamforming performance by providing additional information. However, this benefit comes with the cost of allocating part of the downlink spectrum for pilot transmission, which reduces the SE for data transmission in practice. This trade-off is quantified by multiplying the SE of the integration scheme $\mathcal{R}\left(\mathbf{V}_l^{\mathrm{I}}, \mathbf{H}_l\right)$ by $(1-\rho_l)$, where ρ_l represents the proportion of the pilot overhead for the UE at the lth location as defined in Section III. A. Therefore, the implementation of reduced pilots scheme and its integration with the radio map becomes advantageous only if the actual SE after integration, accounting for the pilot overhead, exceeds the SE achieved by the radio map alone. Mathematically, this condition can be represented as

$$\mathcal{R}\left(\mathbf{V}_l^{\mathrm{I}}, \mathbf{H}_l\right)(1-\rho_l) - \mathcal{R}\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{H}_l\right) > 0. \tag{13}$$

(13) can be readily transformed to (14), which offers a different insight into when it is beneficial to apply reduced pilots and integration schemes.

$$\frac{\mathcal{R}\left(\mathbf{V}_l^{\mathrm{I}}, \mathbf{H}_l\right) - \mathcal{R}\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{H}_l\right)}{\mathcal{R}\left(\mathbf{V}_l^{\mathrm{I}}, \mathbf{H}_l\right)} > \rho_l. \tag{14}$$

The essence of (14) is that the integration scheme only makes sense when the percentage increase in performance of the integration scheme compared to the radio map scheme exceeds the proportion of the pilot overhead. In evaluating the performance of these two schemes for UE at various locations, the performance outcome of the UE at the lth location, denoted by y_l, is labeled as 1 if the integration scheme outperforms the radio map scheme and -1 otherwise, i.e.,

$$y_l = \begin{cases} 1, & \mathcal{R}\left(\mathbf{V}_l^{\mathrm{I}}, \mathbf{H}_l\right)(1-\rho^{l}) > \mathcal{R}\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{H}_l\right) \\ -1, & \text{otherwise.} \end{cases} \tag{15}$$

The labeling strategy in (15) facilitates binary classification of performance outcomes, prompting us to design a discriminator $\mathcal{D}\left(\boldsymbol{g}_l;\theta_{\mathrm{D}}\right)$ to decide which beamforming scheme to adopt based on UE's location, i.e.,

$$\mathbf{V}_l^{\mathrm{SU}} = \begin{cases} \mathbf{V}_l^{\mathrm{I}}, & \mathcal{D}\left(\boldsymbol{g}_l;\theta_{\mathrm{D}}\right) = 1 \\ \mathbf{V}_l^{\mathrm{M}}, & \mathcal{D}\left(\boldsymbol{g}_l;\theta_{\mathrm{D}}\right) = -1, \end{cases} \tag{16}$$

where $\mathbf{V}_l^{\mathrm{SU}}$ is the adopted beamforming vector for the UE at the lth location in single-UE scenario after the discrimination, and θ_{D} is the set of the parameters of the discriminator.

Considering the excellent performance of SVM [43] in binary classification tasks, the discriminator $\mathcal{D}\left(\boldsymbol{g}_l;\theta_{\mathrm{D}}\right)$ is

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

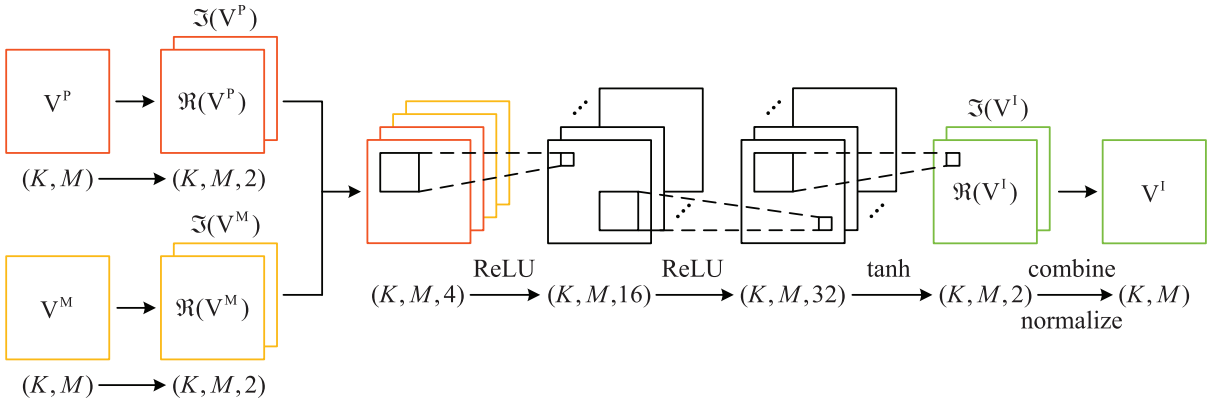


Fig. 6. The architecture of the integration scheme.

designed based on SVM. The objective of SVM is to find a decision boundary (hyperplane) that can not only correctly classify the data points but also maximize the margin between these classes. In the applications of SVM, data points are not perfectly separable by the decision boundary due to outliers. To address this issue, soft margin [44] is implemented which allows for some misclassifications in the training data. The soft-margin SVM solves the optimization problem as

$$\begin{aligned} \min_{\mathbf{a},b,\xi} \quad & \frac{1}{2}||\mathbf{a}||^2 + C\sum_{l=1}^{L}\xi_l \\ s.t. \quad & y_l\big(\mathbf{a}\cdot\phi(\boldsymbol{g}_l)+b\big) \geq 1-\xi_l, \\ & \xi_l \geq 0, \end{aligned} \tag{17}$$

where $\mathbf{a}$ is the normal vector to the hyperplane, b is the bias, C is a constant number that serves as a regularization parameter, allowing control over the penalty assigned to misclassified samples, ξ_l are the slack variables that account for the nonseparability of the data, and $\phi(\boldsymbol{g}_l)$ represents the feature vector obtained after mapping $\boldsymbol{g}_l$ to the high-dimensional space for facilitating the separation.

This classical constrained optimization problem in (17) can be translated into the following dual problem by using the Lagrangian multiplier method [45].

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{L}\alpha_i - \frac{1}{2}\sum_{i=1}^{L}\sum_{j=1}^{L}\alpha_i\alpha_j y_i y_j \phi(\boldsymbol{g}_i)^T\phi(\boldsymbol{g}_j) \\ s.t. \quad & \sum_{i=1}^{L}\alpha_i y_i = 0, \\ & 0 \leq \alpha_i \leq C, \end{aligned} \tag{18}$$

where α_i is the Lagrangian multiplier. It can be seen from (18) that the problem involves calculating $\phi(\boldsymbol{g}_i)^T\phi(\boldsymbol{g}_j)$, which is the inner product of $\boldsymbol{g}_i$ and $\boldsymbol{g}_j$ after being mapped to the high-dimensional space. This process may incur a significant computational cost due to the high dimensionality. To avoid the heavy computation, the Gaussian kernel function, a.k.a. radial basis function (RBF) kernel [46] is introduced to compute the result as follows.

$$\phi(\boldsymbol{g}_i)^T\phi(\boldsymbol{g}_j) = \kappa(\boldsymbol{g}_i,\boldsymbol{g}_j) = \exp\big(-\gamma||\boldsymbol{g}_i-\boldsymbol{g}_j||^2\big), \tag{19}$$

where γ is the kernel parameter inversely proportional to the width of the Gaussian kernel. Then, the problem of (18) can be solved by the sequential minimal optimization (SMO) [47] algorithm in a heuristic way. After solving for α, we can obtain $d(\boldsymbol{g}_l)$ by

$$\begin{aligned} d(\boldsymbol{g}_l) = & \ \mathbf{a}\cdot\phi(\boldsymbol{g}_l)+b \\ = & \sum_{i=1}^{L}\alpha_l y_l \phi(\boldsymbol{g}_i)^T\phi(\boldsymbol{g}_l)+b \\ = & \sum_{i=1}^{L}\alpha_l y_l \kappa(\boldsymbol{g}_i,\boldsymbol{g}_l)+b, \end{aligned} \tag{20}$$

where $d(\boldsymbol{g}_l)$ represents the distance of the point $\phi(\boldsymbol{g}_l)$ to the separating hyperplane. Finally, the designed discriminator can be written as

$$\mathcal{D}(\boldsymbol{g}_l;\theta_{\mathrm{D}}^*) = \begin{cases} 1, & d(\boldsymbol{g}_l) \geq 1 \\ -1, & d(\boldsymbol{g}_l) \leq -1 \end{cases}. \tag{21}$$

Algorithm 2 Discriminator

Input: $\boldsymbol{g}_l \in \mathbb{C}^{2\times 1}$,
Output: $\mathbf{V}_l^{\mathrm{SU}} \in \mathbb{C}^{K\times M}$
1: $\mathbf{V}_l^{\mathrm{M}} \leftarrow \mathcal{G}\left(\boldsymbol{g}_l;\theta_{\mathrm{M}}^*\right)$
2: **if** $\mathcal{D}(\boldsymbol{g}_l;\theta_{\mathrm{D}}^*) = -1$:
3: $\mathbf{V}_l^{\mathrm{SU}} \leftarrow \mathbf{V}_l^{\mathrm{M}}$
4: **else**:
5: transmit reduced pilots $\mathbf{x}_l^{\eta}$ and receive pilots $\boldsymbol{p}_l^{\eta}$
6: $\mathbf{V}_l^{\mathrm{P}} \leftarrow \mathcal{F}\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}^*\right)$
7: $\mathbf{V}_l^{\mathrm{I}} \leftarrow \mathcal{H}\big(\left(\mathbf{V}_l^{\mathrm{M}},\mathbf{V}_l^{\mathrm{P}}\right);\theta_{\mathrm{I}}^*\big)$
8: $\mathbf{V}_l^{\mathrm{SU}} \leftarrow \mathbf{V}_l^{\mathrm{I}}$
Return: $\mathbf{V}_l^{\mathrm{SU}}$

Afterwards, the beamforming scheme can be decided by the discriminator as in (16), and the specific procedure of the SVM-based discriminator for the integration scheme or radio map scheme is summarized in Algorithm 2.

V. Extensions to Multi-UE Scenarios

The aforementioned proposal is only applicable in single-UE scenarios, which can be seen from the loss function in (7). As for the multi-UE scenario with N UEs, considering the interference among UEs, the received signal for the nth UE

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

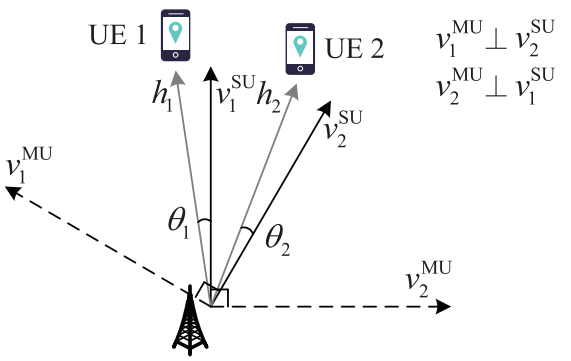


Fig. 7. The diagram of the beamforming vectors for single-UE and multi-UE scenarios.

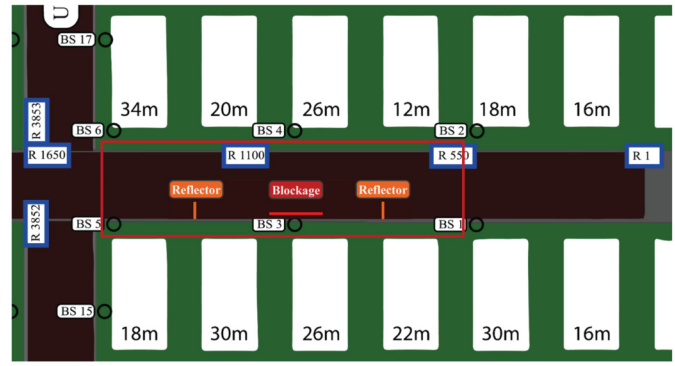


Fig. 8. The top view of the 'O1 Blockage' scenario [49].

over the kth subcarrier should be revised from (2) to account for the interference as

$$y_{n,k} = \mathbf{h}_{n,k}^H \mathbf{x}_{n,k} + \sum_{i=1, i\neq n}^{N} \mathbf{h}_{n,k}^H \mathbf{x}_{i,k} + w_{n,k}$$
$$= \mathbf{h}_{n,k}^H \mathbf{v}_{n,k} s_{n,k} + \sum_{i=1, i\neq n}^{N} \mathbf{h}_{n,k}^H \mathbf{v}_{i,k} s_{i,k} + w_{n,k}, \quad (22)$$

where the second term represents the inter-UE interference. The received signal-to-interference-plus-noise ratio (SINR) of the nth UE over the kth subcarrier is expressed as

$$r_{n,k} = \frac{\left|\mathbf{h}_{n,k}^H \mathbf{v}_{n,k}\right|^2}{\sum_{i=1, i\neq n}^{N} \left|\mathbf{h}_{n,k}^H \mathbf{v}_{i,k}\right|^2 + \sigma^2}. \quad (23)$$

It can be observed from (23) that the inter-UE interference will significantly reduce the SINR, diminishing the downlink SE. To eliminate the interference, a widely employed technique is ZF beamforming [48]. ZF beamforming effectively achieves interference cancellation by designing beamforming vector for the nth UE to be orthogonal to the channels of other UEs, so that the interference is cancelled at the UE side.

However, ZF beamforming relies on access to CSI, which is not available in our end-to-end design. The beamforming vectors are generated directly without the inference of the channel. To address this issue, we adopt the generated beamforming vector to replace the channel information by exploiting the relationship between the beamforming vector $\mathbf{v}^{\mathrm{SU}}$ for single UE and its channel $\mathbf{h}$. In single-UE scenarios, the beamforming vector is obtained by maximizing SE, which essentially equates to maximizing the received signal energy $\left|\mathbf{h}^H \mathbf{v}^{\mathrm{SU}}\right|^2$ under the condition of given noise power. The dot product of vectors $\mathbf{h}$ and $\mathbf{v}^{\mathrm{SU}}$ can be written as

$$\left|\mathbf{h}^H \mathbf{v}^{\mathrm{SU}}\right| = |\mathbf{h}| \left|\mathbf{v}^{\mathrm{SU}}\right| \cos\theta, \quad (24)$$

where θ is the angle between vectors $\mathbf{h}$ and $\mathbf{v}^{\mathrm{SU}}$. When (24) is maximized by the neural network, the angle θ should approach zero and the generated $\mathbf{v}^{\mathrm{SU}}$ should have a direction close to $\mathbf{h}$. Therefore, the problem of finding vectors orthogonal to $\mathbf{h}$ can be transformed into finding vectors orthogonal to $\mathbf{v}^{\mathrm{SU}}$ since they have close directions. As shown in Fig. 7, the beamforming vector $\mathbf{v}_n^{\mathrm{MU}}$ of the nth UE in multi-UE scenario is designed to satisfy $\mathbf{v}_n^{\mathrm{MU}} \perp \mathbf{v}_i^{\mathrm{SU}}$ for $i \neq n$. The beamforming vectors for multi-UE scenarios are generated similarly to the procedures of ZF beamforming. First, the beamforming vectors on the kth subcarrier of all N UEs form the matrix $\mathbf{V}_k^{\mathrm{SU}} = \left[\mathbf{v}_{1,k}^{\mathrm{SU}}, \mathbf{v}_{2,k}^{\mathrm{SU}}, \cdots, \mathbf{v}_{N,k}^{\mathrm{SU}}\right] \in \mathbb{C}^{M\times N}$. Next, the beamforming vectors for multi-UE scenarios are obtained by performing pseudo-inverse on $\mathbf{V}_k^{\mathrm{SU}}$ as

$$\mathbf{V}_k^{\mathrm{MU}H} = \mathbf{V}_k^{\mathrm{SU}\dagger} = \left(\mathbf{V}_k^{\mathrm{SU}H} \mathbf{V}_k^{\mathrm{SU}}\right)^{-1} \mathbf{V}_k^{\mathrm{SU}H}, \quad (25)$$

where $\mathbf{V}_k^{\mathrm{MU}} = \left[\mathbf{v}_{1,k}^{\mathrm{MU}}, \mathbf{v}_{2,k}^{\mathrm{MU}}, \cdots, \mathbf{v}_{N,k}^{\mathrm{MU}}\right] \in \mathbb{C}^{M\times N}$ are the beamforming vectors of all N UEs on the kth subcarrier. After acquiring the multi-UE beamforming vectors of each subcarrier across all the UEs, we can then obtain the beamforming vector of the nth UE in the multi-UE scenario as $\mathbf{V}_n^{\mathrm{MU}} = \left[\mathbf{v}_{n,1}^{\mathrm{MU}}, \mathbf{v}_{n,2}^{\mathrm{MU}}, \cdots, \mathbf{v}_{n,K}^{\mathrm{MU}}\right]^H \in \mathbb{C}^{K\times M}$.

VI. Numerical Simulations and Results

In this section, we present the chosen dataset and provide detailed parameters for numerical simulations. Then, simulation results are illustrated, and discussions are provided.

A. Dataset and Parameters

In the numerical simulations, the DeepMIMO dataset [49] generated by Remcom Wireless InSite [50] is adopted for deterministic propagation channel, while the effects of random scattering are modeled by Rayleigh fading. To make the scenario more representative, the 'O1 Blockage' scenario is chosen and shown in Fig. 8. In this scenario, three surfaces are added to make the channel environment more complex. Specifically, the red line in the figure is the surface to block BS and UE, while two orange lines are the other surfaces acting as reflectors to enable more NLoS paths. The area we select is represented by the red line rectangle in Fig. 8. The BS with index 3 is chosen due to its service quality being significantly affected by the three mentioned surfaces. The range of the locations of UE is from row 501 to row 1500, and the row sampling is 0.1, totalling 100 rows. There are 181 locations in each row, and 20 sites per row are sampled randomly. Therefore, a total of $L = 2000$ locations are sampled.

The communication parameters are set as follows. The frequency f_c is 3.5GHz, and the subcarrier spacing is set to 15kHz. The number of subcarriers is 624, which consists

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

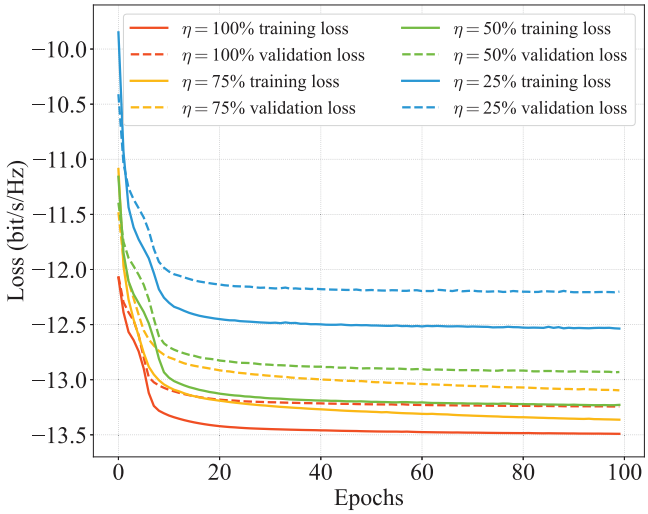


Fig. 9. The training and validation losses of reduced pilots-based beamforming versus epochs.

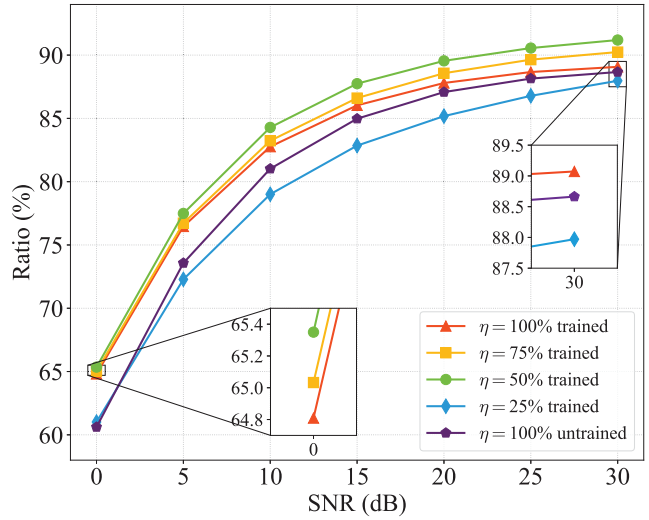


Fig. 10. The performances of the reduced pilots-based beamforming and full pilots versus average SNR.

of 52 resource blocks, and we focus on the first resource block, i.e., $K = 12$. The ratio κ of the power of $\mathbf{H}^r$ to that of $\mathbf{H}^d$ is set to 0.1, indicating that deterministic propagation dominates. We add complex Gaussian noise to the generated CSI, ensuring an average SNR of 30dB across all sampled locations. This approach is closer to practice than the noise-free training data assumption. The positioning error follows a Gaussian distribution with the mean of zero and the standard deviation of σ_e. The uniform linear array (ULA) is applied by the BS for simplification, and the number of transmit antennas M is 32. We consider four pilot configurations with $\eta = 100\%$, 75%, 50%, and 25%, which indicates the ratios of antennas transmitting pilots to all 32 antennas. The corresponding proportion of the pilot overhead ρ for these four configurations are $9.52\%, 7.14\%, 4.76\%$, and 2.38%, respectively.

As for the training, the generated dataset is divided into three distinct subsets: the training set, validation set, and test set, adhering to a ratio of $2 : 2 : 1$, respectively. During the training phase, the training set is used to update the parameters of the neural network, and a monitoring mechanism is implemented where the model exhibiting the lowest loss on the validation set is saved for testing. The network is optimized by the adaptive moment (Adam) [51] optimizer with an adaptive learning rate starting at 1×10^{-3}. If the loss on the validation set fails to reduce over a span of 50 epochs, the learning rate will be decayed by multiplying it by a factor of 0.1. The batch size is set to 128 for training.

B. Performance of Reduced Pilots-Based Beamforming

The neural network for the reduced pilots-based beamforming is trained by (8) and the training and validation losses of the designed four configurations of pilots versus epochs are shown in Fig. 9. All these losses decrease rapidly within the first ten epochs, and for $\eta = 50\%$, 75%, and 100%, the losses stabilize after about 20 epochs. In addition, without considering the decrease in SE caused by pilot overhead, more pilot overhead results in better beamforming performance, which aligns with expectations. Furthermore, the converged losses of $\eta = 50\%$, 75%, and 100% are very close, indicating that further increases in pilot overhead yield little gains in SE.

After taking pilot overhead into consideration, we display the performances of the trained reduced pilots-based beamforming for these four configurations of pilots versus average SNR in Fig. 10. The untrained full pilot configuration is applied as a baseline for comparison. The x-axis represents the average SNR across all sampled locations, while the y-axis depicts the ratio of the SE achieved by the method to the optimal SE. The optimal SE is obtained under the condition of perfect CSI and neglecting pilot overhead. In Fig. 10, the five curves all increase with SNR, where $\eta = 50\%$ achieves the best performance among them, followed by $\eta = 75\%$. This suggests that the concept of reduced pilots is meaningful. Although the accuracy of channel estimation decreases with reduced pilots, the saved time-frequency resources can be used for more valuable data, which makes the actual SE higher than that with full pilots. When $\eta = 100\%$, the performance of the trained network outperforms that of the untrained conventional method especially at low SNR, which indicates that the trained neural network can denoise the input benefiting from the training dataset with high SNR. In particular, the performance of $\eta = 25\%$ is slighter higher than the baseline at 0dB. In summary, $\eta = 50\%$ achieves the best tradeoff between beamforming performance and pilot overhead. Therefore, this pilot configuration will subsequently be adopted for integration with the radio map.

C. Performance of Radio Map-Based Beamforming

According to the training strategy in (10), the training and validation losses of radio map-based beamforming versus epochs are shown in Fig. 11. Both losses decrease rapidly in the initial few epochs and gradually converge to a steady state. After about 800 epochs, the losses no longer show significant fluctuations due to the progressively decreasing learning rate, indicating that the training is almost complete.

To validate the trained radio map, we apply GBM [23], [24], [25] and CKM method [11], [12], [13] as benchmarks.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

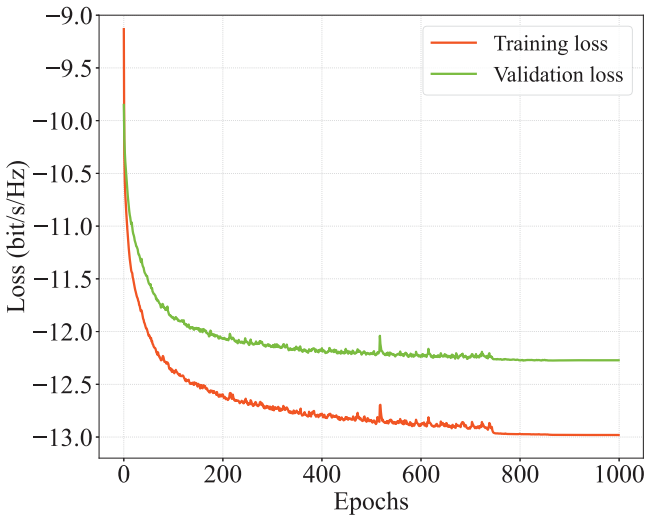


Fig. 11. The training and validation losses of radio map-based beamforming versus epochs.

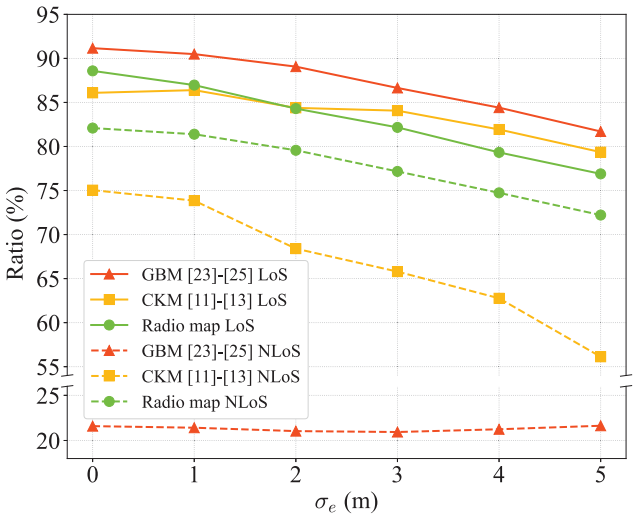


Fig. 12. The performances of GBM, CKM, and radio map under LoS and NLoS conditions versus positioning error.

The performances of these two benchmarks and radio map method under LoS and NLoS conditions versus positioning error are shown in Fig. 12. The x-axis represents the standard deviation of the positioning error, and the y-axis represents the ratio of the SE achieved by the method relative to the optimal SE at 15dB. The radio map is trained using a unified model instead of two separate models for LoS and NLoS conditions because we do not have any inference regarding LoS/NLoS conditions. As shown in Fig. 12, GBM method performs best in LoS condition while worst in NLoS condition, which can only reach approximately 20% of the optimal performance. This is because in LoS condition, the angle calculated based on the positions of BS and UE corresponds to the actual signal transmission angle. However, the situation is quite different in NLoS conditions. The beam is focused on obstacles, which greatly reduces the communication performance. The performance of the radio map approach is comparable to that of CKM, offering better results at lower positioning errors in LoS situations. When $\sigma_e > 2$m, radio map slightly underperforms

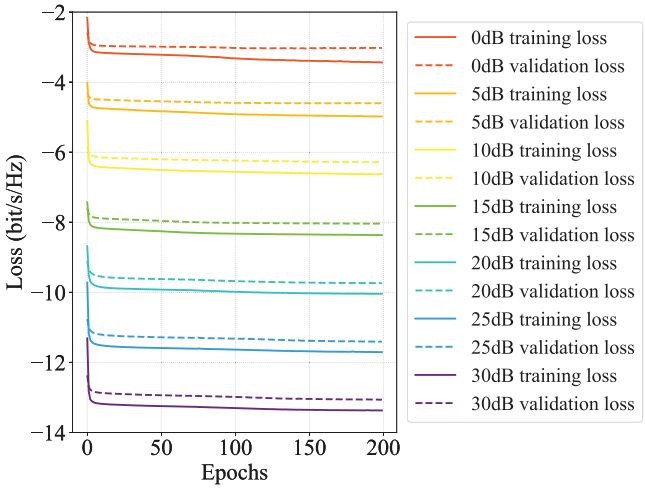


Fig. 13. The training and validation losses of integration scheme versus epochs.

relative to CKM. The deep learning-based radio map approach exhibits lower robustness to positioning errors than traditional methods under LoS conditions. In NLoS situations, radio map is significantly superior to both benchmarks in terms of SE and robustness to positioning errors. This clarifies that deep learning-based methods have a distinct advantage over traditional methods particularly in complex scenarios such as NLoS situations. Taking both LoS and NLoS conditions into consideration, the radio map-based beamforming method is superior to both GBM and CKM methods, exhibiting better applicability in different scenarios.

D. Performance of Integration Scheme for Beamforming

The neural network of the integration scheme is trained based on (12), and the training and validation losses at different average SNRs versus epochs are shown in Fig. 13. The standard deviation of positioning error σ_e is set to 1m for integration. The rapid decrease in loss during the initial epochs, followed by a stable loss later, suggests successful training of the neural network. This indicates that the integration scheme can effectively integrate beamforming vectors generated by radio map and reduced pilots across various SNR levels.

As for the SVM-based discriminator, it is essential to determine the regularization parameter C and kernel parameter γ to achieve optimal classification performance. Considering the potential overfitting issue with the radio map and integration schemes in the training set, SVM training is conducted on the validation set. To evaluate the performance of SVM under different parameter settings, a k-fold cross-validation technique [52] is employed. The k-fold cross-validation involves dividing the dataset into k equal or nearly equal sized folds. The process is repeated k times, with each of the k subsets serving as the test set once and the remaining $k-1$ subsets forming the training set. Here we select $k = 5$. Fig. 14 exhibits the SVM classification accuracies with different combinations of C and γ at 0dB. It can be seen from Fig. 14 that different combinations of parameters yield varying performances. The combination that demonstrates the best performance is chosen for testing where $C = 10$ and $\gamma = 0.01$.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

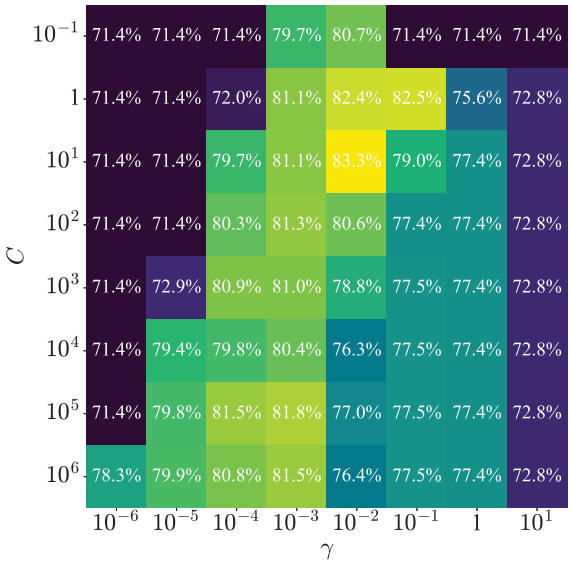


Fig. 14. The heatmap of SVM classification accuracies with different combinations of C and γ at 0dB.

TABLE I

THE TRAINING AND TEST ACCURACIES OF SVM WITH THE BEST PARAMETERS AT DIFFERENT AVERAGE SNRS

Average SNR (dB)	0	5	10	15	20	25	30
C	10	1	100	100	100	100	100
γ	0.01	0.1	0.01	0.01	0.01	0.01	0.01
Training accuracy	83.3%	81.7%	78.5%	78.8%	80.2%	80.8%	78.4%
Test accuracy	84.4%	78.5%	80.3%	82.1%	81.6%	80.3%	78.3%

Table I lists the choice of the best parameters under different average SNRs, along with the associated training and testing accuracies. The parameter C shows significant variation, whereas γ remains relatively constant around 0.01. The closely matched training and test accuracies suggest the excellent generalization ability of the trained SVM. Besides, the accuracy consistently hovers around 80%, affirming the feasibility of discriminating based on UE's locations.

To visually display the results of the SVM-based discriminator, we show the binary classification under three different average SNRs in Fig. 15, where the dots represent the sampled locations of UE. The yellow dots indicate that the radio map approach is better, and green dots suggest that the integration scheme is superior. As shown in Fig. 15, the performances of these two schemes show evident geographic characteristics, which is the basis for the SVM-based discriminator. The classification results of the SVM are represented by corresponding colors. Overall, the accuracy of the classification is quite satisfactory. At 0dB, the radio map method dominates at most locations because the pilots are severely contaminated by noise and thus do not bring significant improvements. As the SNR increases, the benefits provided by pilots gradually become apparent, making the integration scheme surpass the radio map at some locations. However, radio map approach still outperforms integration scheme for many locations even at 30dB. This suggests that at these locations, the performance of the radio map is already sufficient without the need to spare time-frequency resources for pilot transmission.

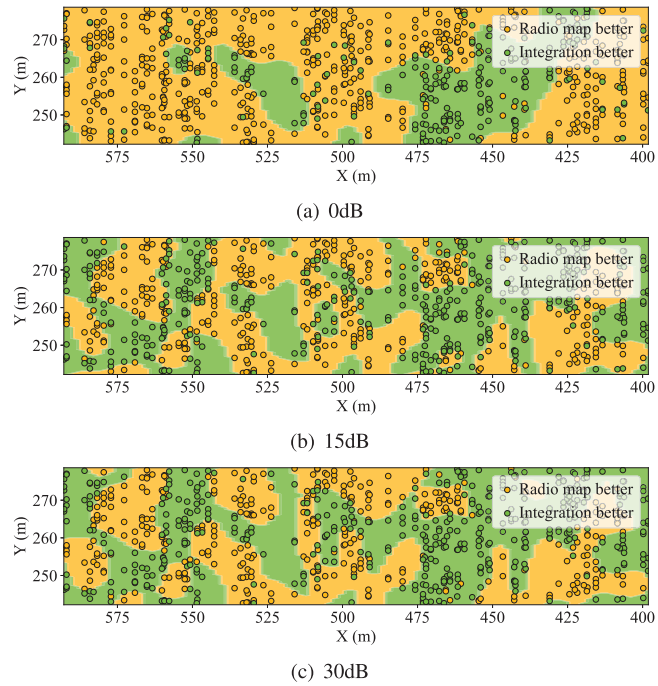


Fig. 15. The classification results of SVM-based discriminator at different average SNRs.

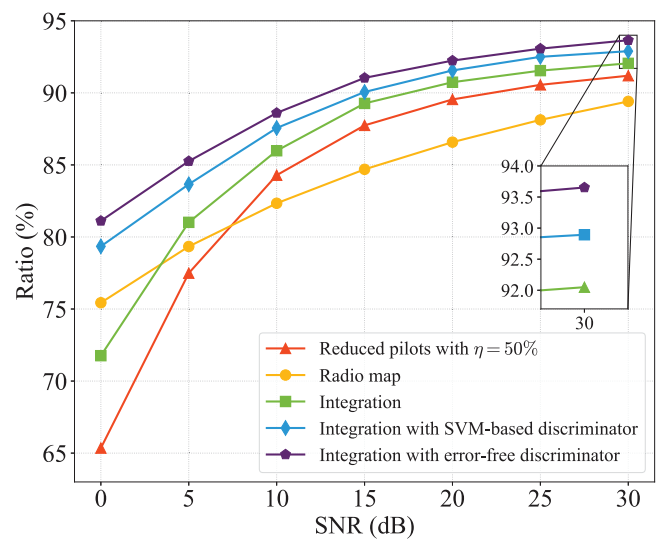


Fig. 16. The performances of integration scheme without and with SVM-based discriminator versus average SNR.

In Fig. 16, we show the performances of integration scheme with and without SVM-based discriminator versus average SNR. To measure the improvement brought by integration scheme, reduced pilots-based beamforming and radio map-based beamforming are applied as baselines for comparison. The integration with error-free discriminator serves as a benchmark to evaluate the performance of the SVM-based discriminator. As shown in Fig. 16, the red curve represents the reduced pilots-based method, which corresponds to the green curve in Fig. 10 and is the best among the four pilot configurations. Compared with radio map, reduced pilots-based method is less robust to SNR, but it outperforms the radio map approach at high SNRs ($\geq$ 10dB). Moreover, the integration

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

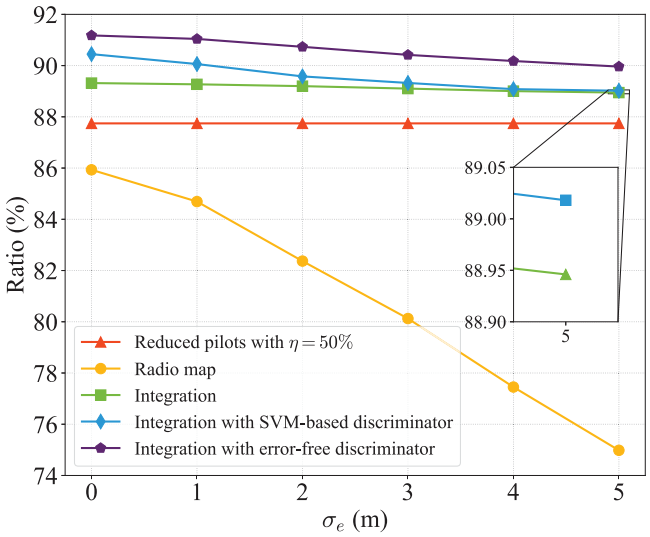


Fig. 17. The performances of integration scheme with and without SVM-based discriminator versus positioning error.

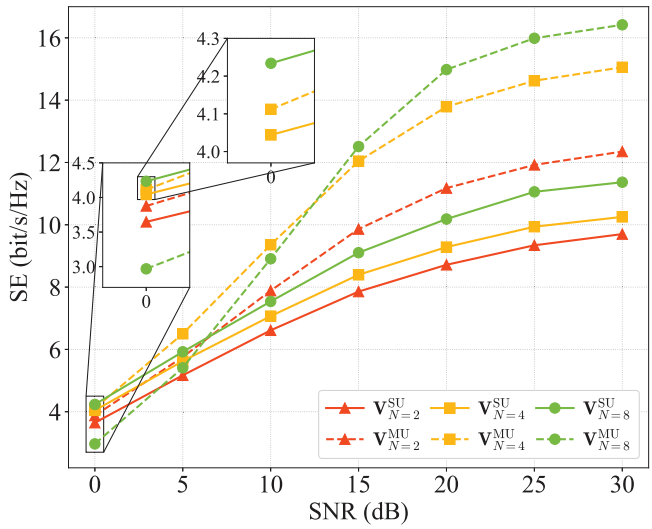


Fig. 18. The SEs of the proposed method in multi-UE scenarios versus average SNR.

scheme without discriminator surpasses both the radio map and reduced pilots schemes at SNRs larger than 5dB. However, the integration scheme falls below the radio map at 0dB. This is because some time-frequency resources are spared for pilots heavily contaminated by noise, which is unworthy. This issue can be effectively addressed by adding a discriminator to the original integration scheme. The SVM-based discriminator can determine which location points benefit from pilots and which do not, further balancing pilot overhead and beamforming performance. In summary, the proposed integration scheme with SVM-based discriminator successfully integrates these two methods, combining the robustness of the radio map with the high performance of pilots at high SNRs.

In Fig. 17, we show the performances of integration scheme with and without SVM-based discriminator versus positioning error at 15dB. It can be seen from the figure that all these methods except for the reduced pilots scheme exhibit a performance decline with increasing positioning error. In particular, radio map demonstrates a pronounced sensitivity to positioning errors, with performance degradation exceeding 10 percent when σ_e reaches 5m. In contrast, reduced pilots-based method relies solely on SNR, and its performance remains unaffected by σ_e when the average SNR maintains at 15dB. This highlights the potential of leveraging pilots to complement the radio map. The performance of the integration scheme remains nearly unchanged, indicating that the proposed scheme effectively integrates the radio map and pilot, thereby maximizing the combined performance of both. With the increase of σ_e, the integration scheme with SVM-based discriminator gradually converges toward the original scheme. This can be attributed to the significant performance degradation of the radio map under large positioning error, prompting the system to allocate part of the downlink time-frequency resources for pilot.

E. Performance in Multi-UE Scenarios

To demonstrate the performance of the proposed scheme in multi-UE scenarios, we study the SE of multi-UE beamforming scheme serving different numbers of UE simultaneously and compare it with the single-UE beamforming scheme in Fig. 18. The standard deviation of positioning error σ_e is 1m for this figure. The x-axis represents the average SNR across all sampled locations. The y-axis stands for the overall SE of all N UE. In multi-UE scenarios, the single-UE beamforming scheme faces interference issues, especially at high SNRs. In comparison, the multi-UE beamforming scheme achieves significant improvements by reducing interference among UE according to the ZF criterion. This is particularly evident as the number of UEs served increases, with larger performance gains observed for $N = 4$ and $N = 8$ in high SNR regions. However, at low SNRs, too many users may lead to the dispersion of BS power, resulting in low received signal strength at the UE side. In such cases, applying ZF to reduce interference might also decrease the signal power, consequently reducing the overall SE. For example, the SE of multi-UE beamforming scheme is lower than that of single-UE beamforming scheme with $N = 8$ at 0dB. This indicates that at low SNRs, directly applying the single-UE beamforming scheme can also be suitable for multi-UE scenarios. Therefore, the choice of service schemes also needs to be flexibly determined based on the number of UE being served simultaneously and the environmental SNR.

F. Analysis of Computational Complexity

The proposed scheme incorporates several machine learning models such as neural networks and SVM. Therefore, it is necessary to analyze the computational complexity for efficiency consideration. Here, we focus on the complexity of the online inference part.

- For the reduced pilots-based beamforming, a CNN is applied, and its complexity is determined by the input dimensionality including the size and channels, the convolution kernel size, and the output dimensionality [53]. The input and the output dimensionalities are both $(K, M, 2)$ and the kernel size is 3×3 so the computational complexity is $\mathcal{O}(K^2M^2)$.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

- The radio map-based beamforming employs an end-to-end DNN. The complexity of the DNN is expressed as the sum of the products of the neurons between all adjacent layers. Since the output dimension $2KM$ is significantly larger than the preceding layers, the overall complexity can be written as $\mathcal{O}(KM)$. For comparison, the computational complexity for GBM is $\mathcal{O}(1)$, which is negligible. The CKM approach involves finding k nearest users for AoA estimation, which requires distance calculation and sorting for N samples. Therefore, the computational complexity is $\mathcal{O}(N \log N)$.
- The integration scheme also applied a CNN and its complexity is $\mathcal{O}(K^2M^2)$, which is the same as that of the reduced pilots-based beamforming.
- For the SVM, the computational complexity is related to the number of support vectors and the input dimensionality [43]. Given the non-linear separability of the data, the number of support vectors is approximate to the number of location samples N. The input consists of two-dimensional coordinates, so the computational complexity is $\mathcal{O}(N)$.

A core component of our proposed scheme is the use of SVM to determine the necessity of pilots. When pilots are not introduced, the computational complexity is $\mathcal{O}(N+KM)$ while $\mathcal{O}(N+KM+K^2M^2)$ otherwise. It is evident that the majority of the computational complexity in our scheme arises from CNN. The application of the SVM-based discriminator can avoid unnecessary time-frequency resources and computational cost for pilots. In other words, the SVM ensures that the resources allocated for pilot are worthwhile and contribute to performance improvement.

VII. Conclusion

In this paper, we first propose a reduced pilots-based beamforming method and investigate the trade-off between pilot overhead and beamforming performance from the perspective of SE. Then, we analyze the performances of the proposed reduced pilots method and radio map method. According to the distinct characteristics of these two approaches, an integration scheme is proposed to combine their advantages and compensate for shortcomings. To avoid unnecessary pilot overhead, a discriminator based on SVM is designed, which can choose whether reduced pilots are needed to enhance performance based on locations. The proposed scheme in this paper effectively integrates radio map and reduced pilots, achieving a balance between pilot overhead and beamforming performance, and serving as a comprehensive solution that adapts to varying environments.

References

[1] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive MIMO for next generation wireless systems," *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 186–195, Feb. 2014.

[2] F. Rusek et al., "Scaling up MIMO: Opportunities and challenges with very large arrays," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 40–60, Jan. 2012.

[3] L. Lu, G. Y. Li, A. L. Swindlehurst, A. Ashikhmin, and R. Zhang, "An overview of massive MIMO: Benefits and challenges," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 742–758, Oct. 2014.

[4] H. Yang and T. L. Marzetta, "Performance of conjugate and zero-forcing beamforming in large-scale antenna systems," *IEEE J. Sel. Areas Commun.*, vol. 31, no. 2, pp. 172–179, Feb. 2013.

[5] *NR; Physical Channels and Modulation*, document TS 38.211, 3GPP, Jan. 2024. [Online]. Available: https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3213

[6] *NR; Physical Layer Procedures for Data*, document TS 38.214, 3GPP, Jan. 2024. [Online]. Available: https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3216

[7] C. Wen, W. Shih, and S. Jin, "Deep learning for massive MIMO CSI feedback," *IEEE Wireless Commun. Lett.*, vol. 7, no. 5, pp. 748–751, Oct. 2018.

[8] J. Jose, A. Ashikhmin, T. L. Marzetta, and S. Vishwanath, "Pilot contamination and precoding in multi-cell TDD systems," *IEEE Trans. Wireless Commun.*, vol. 10, no. 8, pp. 2640–2651, Aug. 2011.

[9] *Framework and Overall Objectives of the Future Development of IMT for 2030 and Beyond*, document ITU-R WP5D, ITU-R, Jun. 2023. [Online]. Available: https://www.itu.int/md/R19-WP5D-230612-TD-0905

[10] S. Bi, J. Lyu, Z. Ding, and R. Zhang, "Engineering radio maps for wireless resource management," *IEEE Wireless Commun.*, vol. 26, no. 2, pp. 133–141, Apr. 2019.

[11] Y. Zeng and X. Xu, "Toward environment-aware 6G communications via channel knowledge map," *IEEE Wireless Commun.*, vol. 28, no. 3, pp. 84–91, Jun. 2021.

[12] D. Ding, D. Wu, Y. Zeng, S. Jin, and R. Zhang, "Environment-aware beam selection for IRS-aided communication with channel knowledge map," in *Proc. IEEE Globecom Workshops*, Dec. 2021, pp. 1–6.

[13] D. Wu, Y. Zeng, S. Jin, and R. Zhang, "Environment-aware and training-free beam alignment for mmWave massive MIMO via channel knowledge map," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Jun. 2021, pp. 1–7.

[14] L. Le Magoarou, T. Yassine, S. Paquelet, and M. Crussière, "Deep learning for location based beamforming with NLOS channels," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 8812–8816.

[15] W. Wang, B. Yang, and W. Zhang, "Deep learning-based radio map for MIMO-OFDM downlink precoding," *J. Commun. Inf. Netw.*, vol. 8, no. 3, pp. 203–211, Sep. 2023.

[16] B. Yang, W. Wang, and W. Zhang, "Cell-free massive MIMO beamforming based on radio map," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2024, pp. 4506–4511.

[17] A. Mayouche, A. Metref, and J. Choi, "Downlink training overhead reduction technique for FDD massive MIMO systems," *IEEE Signal Process. Lett.*, vol. 25, no. 8, pp. 1201–1205, Aug. 2018.

[18] M. B. Mashhadi and D. Gündüz, "Pruning the pilots: Deep learning-based pilot design and channel estimation for MIMO-OFDM systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6315–6328, Oct. 2021.

[19] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.

[20] S. Zhang, Y. Liu, F. Gao, C. Xing, J. An, and O. A. Dobre, "Deep learning based channel extrapolation for large-scale antenna systems: Opportunities, challenges and solutions," *IEEE Wireless Commun.*, vol. 28, no. 6, pp. 160–167, Dec. 2021.

[21] T. Raviv, S. Park, O. Simeone, Y. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.

[22] T. Raviv and N. Shlezinger, "Data augmentation for deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 8259–8274, Nov. 2023.

[23] C. Liu and R. Malaney, "Location-based beamforming and physical layer security in Rician wiretap channels," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7847–7857, Nov. 2016.

[24] P. Kela et al., "Location based beamforming in 5G ultra-dense networks," in *Proc. IEEE 84th Veh. Technol. Conf. (VTC-Fall)*, Sep. 2016, pp. 1–7.

[25] W. Wang and W. Zhang, "Joint beam training and positioning for intelligent reflecting surfaces assisted millimeter wave communications," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6282–6297, Oct. 2021.

[26] E. J. Candes and T. Tao, "Decoding by linear programming," *IEEE Trans. Inf. Theory*, vol. 51, no. 12, pp. 4203–4215, Dec. 2005.

[27] W. Dai and O. Milenkovic, "Subspace pursuit for compressive sensing signal reconstruction," *IEEE Trans. Inf. Theory.*, vol. 55, no. 5, pp. 2230–2249, May 2009.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.

[28] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.

[29] A. Garcia-Rodriguez and C. Masouros, "Exploiting the tolerance of massive MIMO to incomplete CSI for low-complexity transmission," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2015, pp. 1–6.

[30] R. O. Adeogun, P. D. Teal, and P. A. Dmochowski, "Extrapolation of MIMO mobile-to-mobile wireless channels using parametric-model-based prediction," *IEEE Trans. Veh. Technol.*, vol. 64, no. 10, pp. 4487–4498, Oct. 2015.

[31] W. Alves, I. Correa, N. González-Prelcic, and A. Klautau, "Deep transfer learning for site-specific channel estimation in low-resolution mmWave MIMO," *IEEE Wireless Commun. Lett.*, vol. 10, no. 7, pp. 1424–1428, Jul. 2021.

[32] O. Wang, J. Gao, and G. Y. Li, "Learn to adapt to new environments from past experience and few pilot blocks," *IEEE Trans. Cognit. Commun. Netw.*, vol. 9, no. 2, pp. 373–385, Apr. 2023.

[33] S. Park, H. Jang, O. Simeone, and J. Kang, "Learning to demodulate from few pilots via offline and online meta-learning," *IEEE Trans. Signal Process.*, vol. 69, pp. 226–239, 2021.

[34] L. Zhu, C. Bockelmann, T. Schier, S. E. Hajri, and A. Dekorsy, "Enhancing CNN-based channel estimation using transfer learning in OFDM systems," in *Proc. IEEE 24th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Sep. 2023, pp. 21–25.

[35] S. Zhang, X. Li, M. Zong, X. Zhu, and R. Wang, "Efficient kNN classification with different numbers of nearest neighbors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 5, pp. 1774–1785, May 2018.

[36] G. Y. Lu and D. W. Wong, "An adaptive inverse-distance weighting spatial interpolation technique," *Comput. Geosci.*, vol. 34, no. 9, pp. 1044–1055, 2008.

[37] S. Coleri, M. Ergen, A. Puri, and A. Bahai, "Channel estimation techniques based on pilot arrangement in OFDM systems," *IEEE Trans. Broadcast.*, vol. 48, no. 3, pp. 223–229, Sep. 2002.

[38] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[39] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn.*, 2010, pp. 807–814.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[41] H.-S. Tan and J. Huang, "DGPS-based vehicle-to-vehicle cooperative collision warning: Engineering feasibility viewpoints," *IEEE Trans. Intell. Transp. Syst.*, vol. 7, no. 4, pp. 415–428, Dec. 2006.

[42] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, 1989.

[43] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Their Appl.*, vol. 13, no. 4, pp. 18–28, Jul. 1998.

[44] J. Shawe-Taylor and N. Cristianini, "On the generalization of soft margin algorithms," *IEEE Trans. Inf. Theory*, vol. 48, no. 10, pp. 2721–2735, Oct. 2002.

[45] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[46] B. Schölkopf et al., "Comparing support vector machines with Gaussian kernels to radial basis function classifiers," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2758–2765, Nov. 1997.

[47] J. Platt, "Sequential minimal optimization: A fast algorithm for training support vector machines," Microsoft, Redmond, WC, USA, Tech. Rep. MSR-TR-98-14, Apr. 1998.

[48] Q. H. Spencer, A. L. Swindlehurst, and M. Haardt, "Zero-forcing methods for downlink spatial multiplexing in multiuser MIMO channels," *IEEE Trans. Signal Process.*, vol. 52, no. 2, pp. 461–471, Feb. 2004.

[49] A. Alkhateeb, "DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications," in *Proc. Inf. Theory Appl. Workshop (ITA)*, San Diego, CA, USA, Feb. 2019, pp. 1–8.

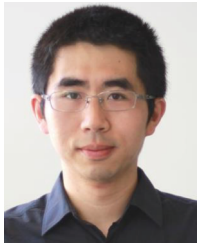
[50] Remcom.*Wireless InSite*. Accessed: Apr. 2024. [Online]. Available: http://www.remcom.com/wireless-insite

[51] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[52] J. D. Rodriguez, A. Perez, and J. A. Lozano, "Sensitivity analysis of k-fold cross validation in prediction error estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 3, pp. 569–575, Mar. 2010.

[53] P. Molchanov, S. Tyree, T. Karras, T. Aila, and J. Kautz, "Pruning convolutional neural networks for resource efficient inference," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–17.

Bin Yang (Graduate Student Member, IEEE) received the B.S. degree in telecommunication engineering from Nanjing University of Posts and Telecommunications, Nanjing, China, in 2018, and the M.S. degree in communications and information systems from Xidian University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree with The University of New South Wales, Sydney, Australia. His research interests include radio map, wireless communications, and machine learning.



Wei Wang (Senior Member, IEEE) received the B.S. degree from Central South University, Changsha, China, in 2010, the M.S. degree from Southeast University, Nanjing, China, in 2013, and the Ph.D. degree from The University of New South Wales, Sydney, Australia, in 2017.

From 2018 to 2021, he was a Post-Doctoral Research Fellow with The University of New South Wales. He is currently a Senior Research Scientist (equivalent to Associate Professor) with the Peng Cheng Laboratory, Shenzhen, China. His research interests include millimeter-wave communications, machine learning for wireless communications, and cyber security.

Dr. Wang was a recipient of the 2023 IEEE ComSoc AP Outstanding Young Researcher Award and the Best Paper Award in the IEEE ICCC 2016 and 2024. He served as an Editor for IEEE WIRELESS COMMUNICATIONS LETTERS. He also served as the Co-Chair for the Signal Processing for Communications Symposium of IEEE ICCC 2021, the AI Enabled Wireless and Networking Symposium of IEEE ICCC 2024, the Convergence of Sensing, Communication, and Computing Symposium of IEEE WCSP 2024, and the Workshops on "Radio Map for 6G Communications" of IEEE VTC2024-Spring and IEEE ICCC 2025.

Wei Zhang (Fellow, IEEE) received the Ph.D. degree from The Chinese University of Hong Kong in 2005. He is currently a Professor with the School of Electrical Engineering and Telecommunications, The University of New South Wales, Sydney, Australia. His current research interests include UAV communications, 5G, and beyond. He has served as a member for various ComSoc boards/standing committees, including Journals Board, Technical Committee Recertification Committee, Finance Standing Committee, Information Technology Committee, and Steering Committee for IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING and IEEE NETWORKING LETTERS. He received six best paper awards from the IEEE conferences and ComSoc technical committees. He also serves as an Area Editor for IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS and the Editor-in-Chief for JOURNAL OF COMMUNICATIONS AND INFORMATION NETWORKS. Previously, he served as an Editor for IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING, and IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS COGNITIVE RADIO SERIES. Within the IEEE ComSoc, he has taken many leadership positions, including the Member-at-Large on the Board of Governors from 2018 to 2020, the Chair of the Wireless Communications Technical Committee from 2019 to 2020, the Vice Director of the Asia-Pacific Board from 2016 to 2021, the Editor-in-Chief of IEEE WIRELESS COMMUNICATIONS LETTERS from 2016 to 2019, the Technical Program Committee Chair of APCC 2017 and ICCC 2019, and the Award Committee Chair of the Asia-Pacific Board and the Technical Committee on Cognitive Networks. He was an IEEE ComSoc Distinguished Lecturer from 2016 to 2017. He is the Vice President of the IEEE Communications Society.

Authorized licensed use limited to: University of New South Wales. Downloaded on April 23,2026 at 04:51:47 UTC from IEEE Xplore. Restrictions apply.